

ChE453: Capstone Project
Combined bi-weekly reports: 4, 5, 6
Group 4

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Reactor Design:

Heat duty	-285.445205	kW
Reactor temperature		
Minimum	63.4	C
Maximum	63.4	C
Residence time	2.56693	hr
Thermal fluid inlet		
Temperature		
Vapor fraction		

Reactor dimensions		
Length	40	meter
Diameter	8	meter
Elevation		
☒ Reactor rise	0	meter
☐ Reactor angle	0	rad
Valid phases		
Process stream	Vapor-Liquid	
Thermal fluid stream	Vapor-Liquid	2nd Liquid

► - Mole Flows	kmol/hr	18100	18100
► MEB	kmol/hr	0.0578281	57.8279
► BENZA	kmol/hr	99.7273	41.9572
► WATER	kmol/hr	0.0572491	57.8274
► METH	kmol/hr	17999.9	17942.2
► PTH	kmol/hr	0.214915	0.214915
► OXYGEN	kmol/hr	0	0
► NITROGEN	kmol/hr	0	0
► + Mole Fractions			
► + Mass Flows	kg/hr	588981	588981
► + Mass Fractions			
► Volume Flow	cum/sec	0.217557	0.217592

Flow Regime:

Average Velocity: **0.004328 m/s**

Reynolds Number: **36960.81**

Flow Regime: **Turbulent** ($Re \geq 4000$)

The flow regime obtained in our reactor is turbulent which is consistent with standard industrial practices. We usually operate PFR in turbulent regime due to following reasons:

- Complete radial mixing (uniform temperature/concentration)
- High heat transfer coefficients
- Minimized axial dispersion (plug-flow approach)

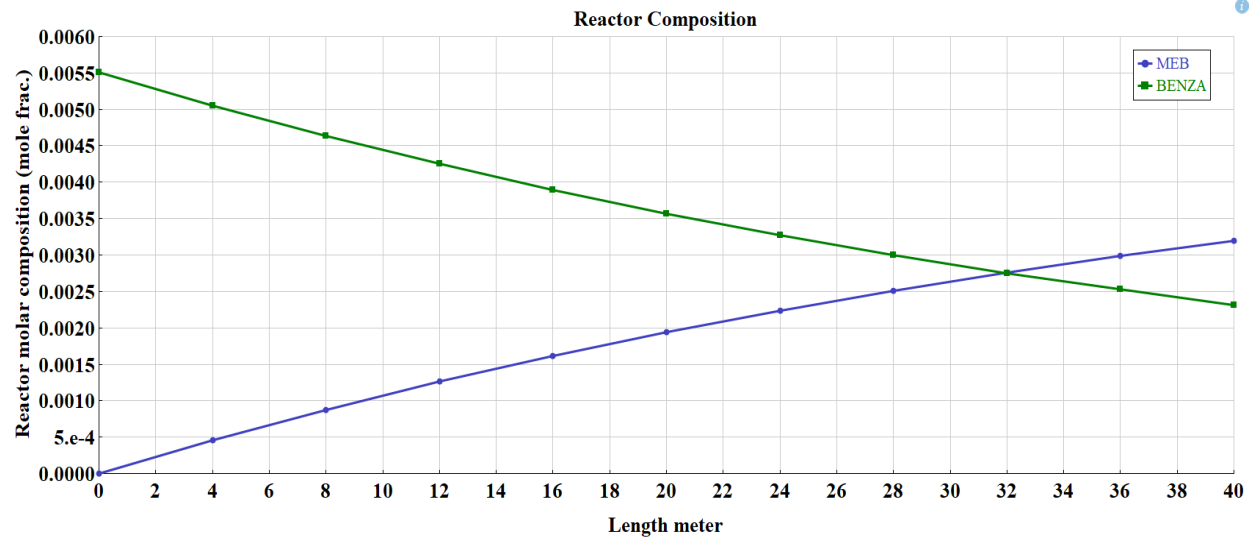
Residence time:

The residence time obtained in our reactor is **2.56 hours**, which is consistent with typical industrial ranges (**1–5 hours**) for plug flow reactors, especially for reactions requiring extended contact time.

There exists an inherent tradeoff between reaction conversion and residence time in plug flow reactors. Increasing reactor volume typically enhances conversion due to prolonged reactant contact time; however, this also leads to longer residence times, which may impact operational efficiency.

In our system, we obtained a **single-pass conversion of 58%**. We implement **reactant recycling** to optimize resource utilization and minimize waste, allowing unreacted species to re-enter the reactor. This significantly boosts the **overall conversion to 99.628%**.

Composition Profile of Reactor:



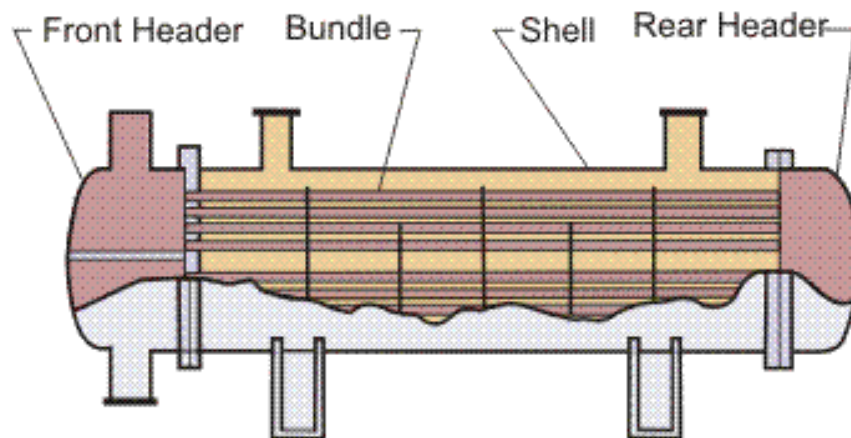
Fresh methanol feed rate = **61.3666 kmol/hr**

Fresh benzoic acid feed rate = **57.812 kmol/hr**

Methyl Benzoate outlet flow rate = **57.77 kmol/hr**

Detailed Heat Exchanger:

The Tubular Exchanger Manufacturers Association (TEMA) is an association of manufacturers of shell and tube heat exchangers. TEMA has established a set of construction standards for Shell and Tube Heat Exchangers.



E-Type shell

This is the most commonly used shell type, suitable for most duties and applications. Other shell types only tend to be used for special duties or applications.

Tube diameter layout and pitch

Tubes may range in diameter from 12.7 mm (0.5 in) to 50.8 mm (2 in), but 19.05 mm (0.75 in) and 25.4 mm (1 in) are the most common sizes.

The tubes are laid out in triangular or square patterns in the tube sheets.

We are taking the outer diameter 20mm and inner diameter 16mm of the tube.

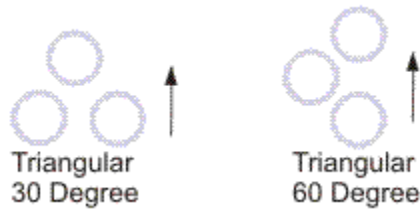
Tube Specifications:

- Outer diameter: 20 mm
- Inner diameter: 16 mm
- Standard Length: 4.88 m
- Area of tube (πDL) = 0.303324 m²
- Number of tubes: 2425
- $T_{in} = 64.2\text{ }^{\circ}\text{C}$, $T_{out} = 35\text{ }^{\circ}\text{C}$
- $T_{mean} = 49.6\text{ }^{\circ}\text{C}$
- Single pass
- Mass Flow rate(kg/hr) = 382130.9
- Mass Flow rate (kg/s) = 106.147
- Mass Flow velocity = 217.815 kg/m²s
- Linear velocity = 0.275 m/s

Pitch: The tube pitch is the shortest center-to-center distance between tubes. The tube spacing is given by the tube pitch/tube diameter ratio, which is normally 1.25 or 1.33.

Pitch = 1.25 * outer diameter = 25 mm

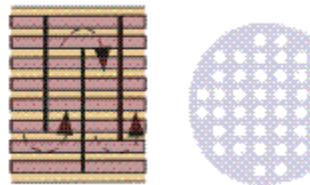
The triangular arrangement allows more tubes in a given space. In our system the number of tubes is 2425 so we'll use triangular arrangement.



Baffle types

Baffles are installed on the shell side to give a higher heat-transfer rate due to increased turbulence and to support the tubes thus reducing the chance of damage due to vibration.

- *Single segmental* (this is the most common),



Baffles Specification:

- Number of baffles = 5
- Baffle spacing(20 % of shell diameter) = 288.25 mm

Shell Specification:

- Diameter = 1441 mm
- Mass flow rate = 600000 kg/hr
- Mass flow rate = 166.67 kg/s
- Mass flow velocity = 2005.78 kg/m²s
- T_{in} = 25 °C
- T_{out} = 40 °C

- $T_{\text{mean}} = 32.5\text{ }^{\circ}\text{C}$
- Reynold's number = 32004.6
- Prandtl number = 6.2
- Baffle cut = 25%

Calculations from Excel Sheet:

Cph (kJ/kgK)	mh (kg/hr)	Thout	Thin	Tcout	Tcin	mc (kg/hr)	Cpc (kJ/kgK)
3.37	382131	35	64.2	40	25	600000	4.18

Results of Heat Exchanger from Aspen from shortcut method:

Main Flowsheet × HX2 (HeatX) - Setup × Results Summary - Run Status × HX2 (HeatX) - Thermal Results ×

Summary Balance Exchanger Details Pres Drop/Velocities Zones Utility Usage Status

Heatx results

Calculation Model:

	Inlet		Outlet	
Hot stream:	D2		METHOUT	
Temperature	64.2007	C	35	C
Pressure	1	bar	1	bar
Vapor fraction	0		0	
1st liquid / Total liquid	1		1	
Cold stream	METHCW		RNKWATOU	
Temperature	25	C	39.4416	C
Pressure	1	bar	1	bar
Vapor fraction	0		0	
1st liquid / Total liquid	1		1	
Heat duty	9983.2	kW		

Exchanger details		
▶	Calculated heat duty	9983.2 kW
▶	Required exchanger area	721.456 sqm
▶	Actual exchanger area	721.456 sqm
▶	Percent over (under) design	0
▶	Average U (Dirty)	0.85 kW/sqm-K
▶	Average U (Clean)	
▶	UA	613.238 kW/K
▶	LMTD (Corrected)	16.2795 C
▶	LMTD correction factor	1
▶	Thermal effectiveness	
▶	Number of transfer units	
▶	Number of shells in series	1
▶	Number of shells in parallel	

Calculated:

- Heat exchanger area = 735.3 m²
- UA value = 625 kW/K

Aspen Results:

- Heat exchanger area = 721.45 m²
- UA value = 613.24 kW/K

The above calculated data matches with results obtained from Aspen:

Exchanger Design and Rating:

EDR is a feature in Aspen used for the thermal and mechanical design and analysis of heat exchangers.

Design specifications of EDR:

✓ Design Specifications

Codes and Standards

Design Code	ASME Code Sec VIII Div 1
Service class	Normal
TEMA class	B - chemical service
Material standard	ASME
Dimensional standard	ANSI - American
Use BPVC VIII Div 2 Class 1 or Class 2	No

Design Conditions

		Tube Side Hot Side	Shell Side Cold Side
Design pressure (gauge)	bar	3	3
Design temperature	°C	100	75
Minimum design metal temperature	°C		
Vacuum design pressure (gauge)	psi		
Test pressure (gauge)	psi		
Corrosion allowance	in	0	0
Radiography		Spot	Spot

Tube material used:

✓ Vessel Materials

✓ Cladding/Gasket Materials

✓ Tube Properties

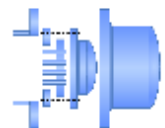
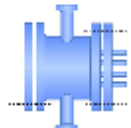
Default exchanger material	CuNi 90/10	26
Component	Material	Designator
Cylinder - hot side	CuNi 90/10	26
Cylinder - cold side	CuNi 90/10	26
Tubesheet	CuNi 90/10	26
Double tubesheet (inner)	Set Default	0
Baffles	CuNi 90/10	26
Tube material	CuNi 90/10	26
Fin material	Set Default	0

Final tube specifications:

✓ Tube	Lowfins	Fins	✓ Inserts	Internal Enhancements
Number of tubes (total)	356			
Number of tubes plugged				
Tube length	118.1102	in		
Tube type	Plain			
Tube outside diameter	20	mm		
Tube wall thickness	2	mm		
Wall specification	Average			
Tube pitch	25	mm		
Tube pattern	30-Triangular			
Tube material	CuNi 90/10	26		
Tube surface	Smooth			
Tube wall roughness		in		
Tube cut angle (degrees)				



✓ Shell/Heads	✓ Covers	✓ Tubesheets	✓ Flanges
---------------	----------	--------------	-----------



Front head type	A - channel & removable cover
Shell type	E - one pass shell
Rear head type	S - floating head with backing device
Exchanger position	Horizontal
Location of front head for vertical units	Set default
"E" shell flow direction (inlet nozzle location)	Near rear head
Double pipe or hairpin unit shell pitch	
Tubeside inlet at front head	Set default
Flow within multi-tube hairpin (M-shell)	Set default
Overall flow for multiple shells	Countercurrent

Heat Exchanger Network Design using Pinch Technology:

Minimum Approach Temperature Chosen: 10 °C

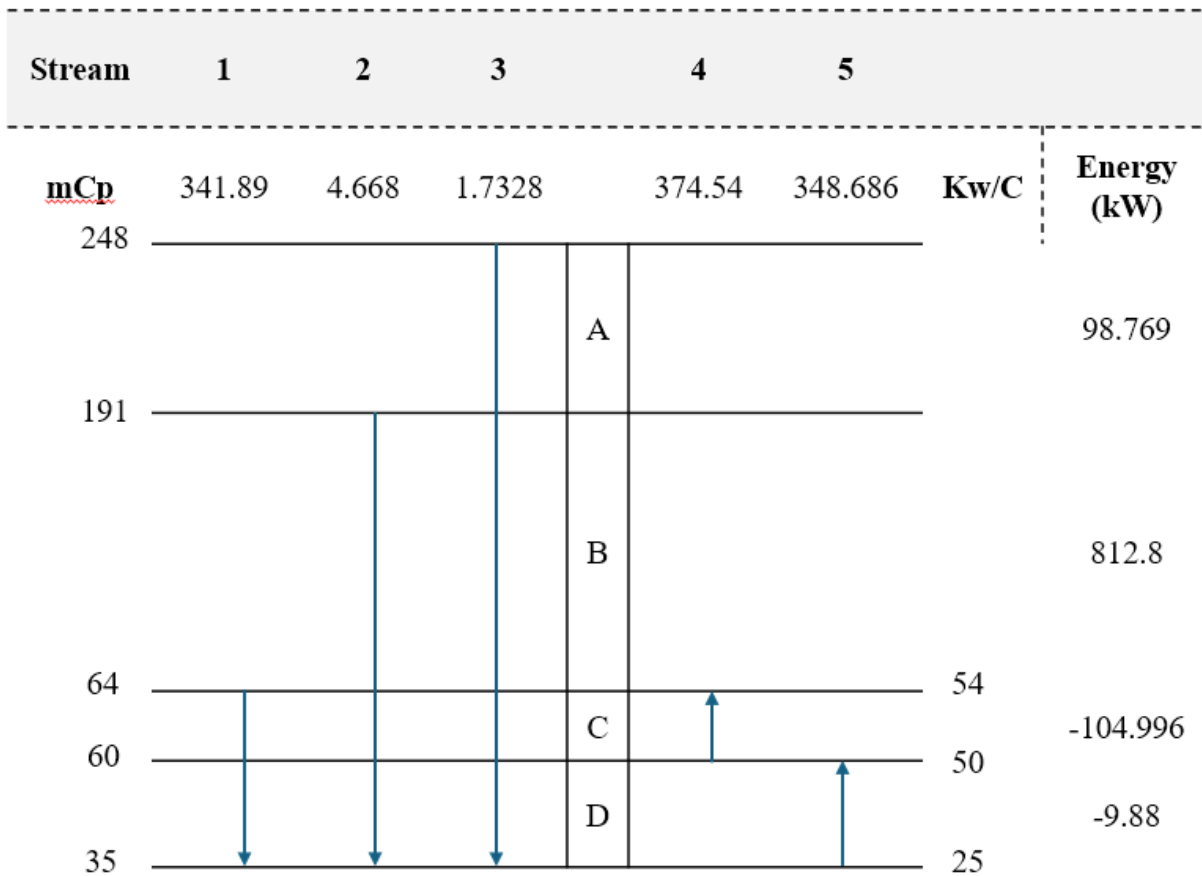
Process Stream Involved: -

Stream	Condition	Flowrate (kg/s)	mCp (kW/C)	T _{in}	T _{out}	Q _{av} (kW)
1 (D2)	Hot	106.147	341.89	64	35	9914.81
2 (D3)	Hot	2.71539	4.668	191	35	728.208
3 (D4)	Hot	0.9502	1.7328	248	35	369.0864
4 (LIQOUT)	Cold	110.203	374.54	50	54	1498.16
5 (MIXOUT)	Cold	110.203	348.686	25	50	8717.15

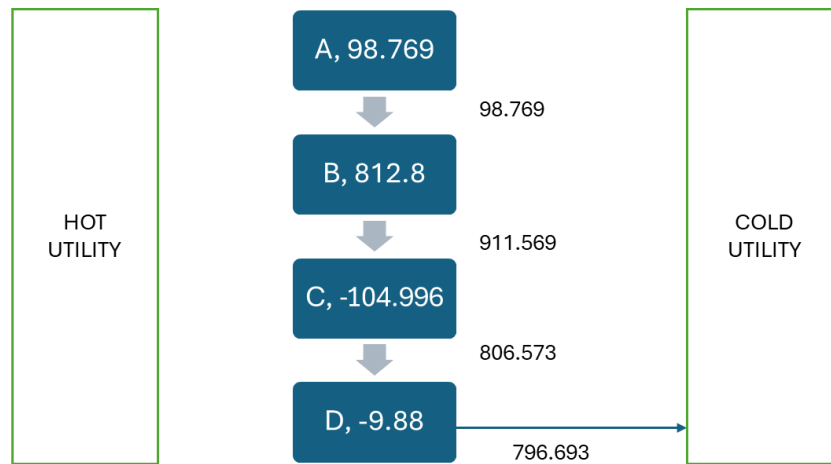
The above table lists a total of 5 process streams involved in the MUMNE design. However we had to conduct the reaction at around 50 °C, which is well suited for the liquid phase reaction we are performing. But we do not have sufficient data at this stage regarding temp dependency of the reaction kinetics, which we are trying to figure out.

Temperature Interval Diagram: -

The temperature interval diagram of the five process streams. The ones to the left are hot streams meant to be cooled. Their energy requirements have been taken positive. Reverse has been taken for the streams to the right, which are meant to be heated. The energy written next to each block is the sum of net required energy for all streams across that block.

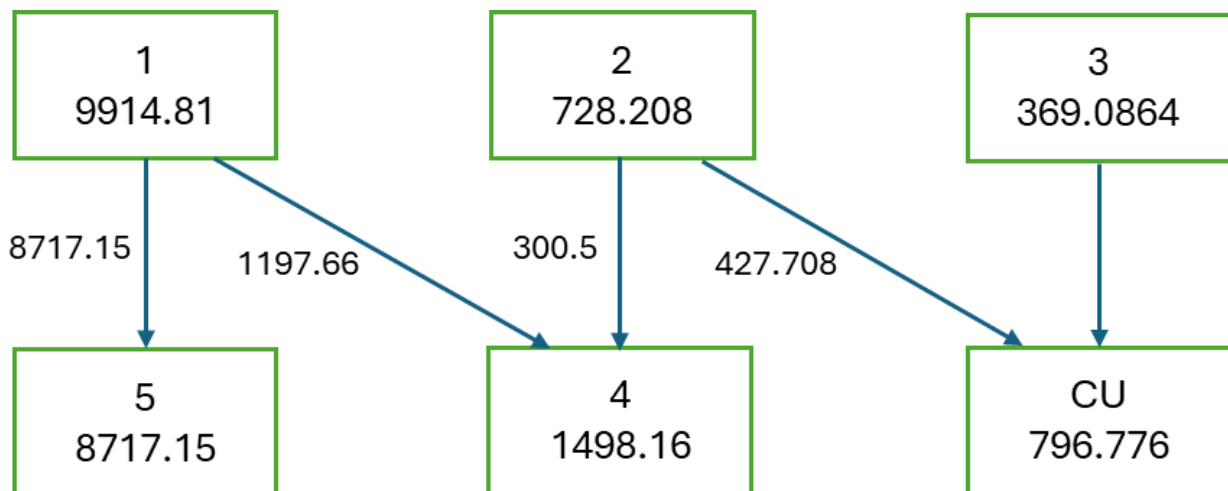


Cascade Diagram: -



As per the above cascade diagram, it can be noticed that a net excess of energy occurs while cascading downwards. Hence, the process involves **no pinch**, and all the excess heat will be transferred to the cold utility.

Minimum Number of Heat Exchangers: -

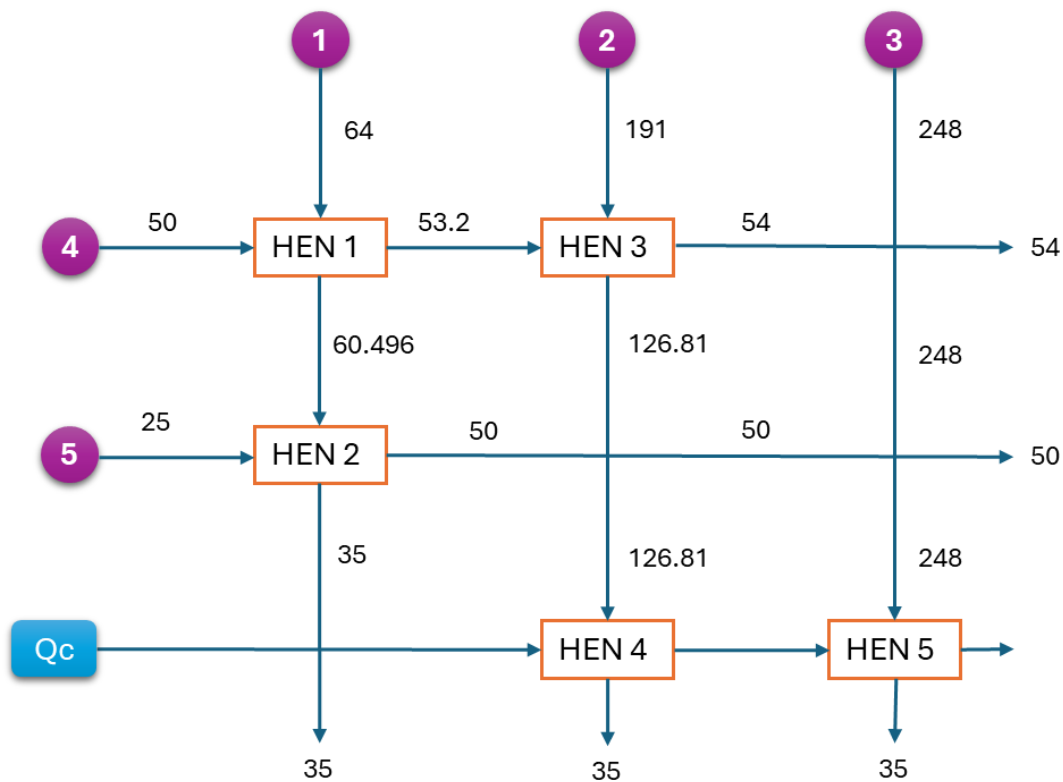


Since no exact matchups could be found,
Minimum number of Heat Exchanger = No of process stream + Utility Stream - 1 = 5 which is same as number of arrows in the above

Note that the upper blocks represent hot process streams and lower blocks, the cold process streams. CU represents cold utility.

Heat Exchanger Network Design: -

The streams were matched according to: $m \cdot cp_{hot} \leq m \cdot cp_{cold}$, starting from the ones with lower temperature and moving upwards. The design is not according to the MUMNE network above as minimum approach temperature has been considered here.

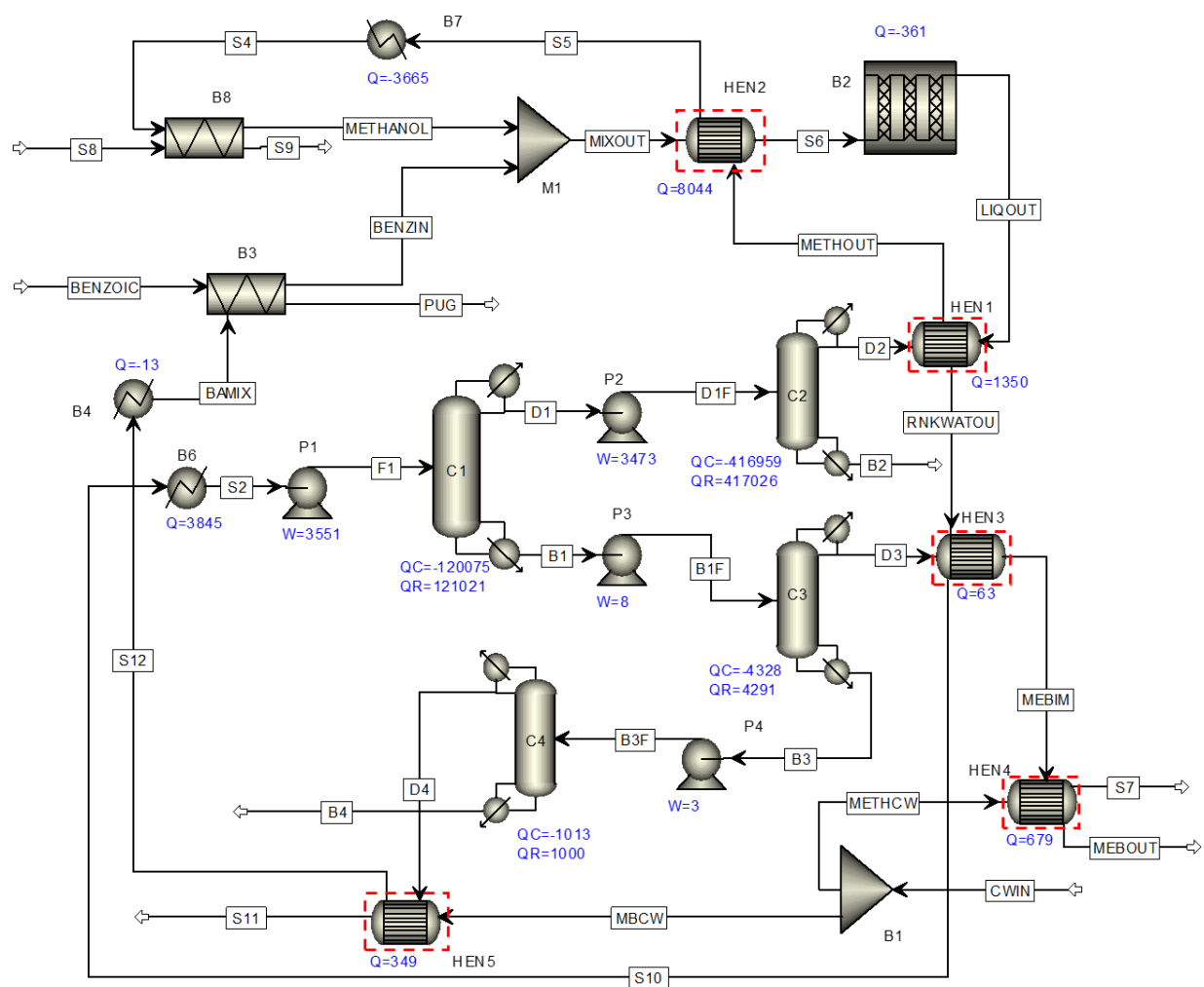


Legend



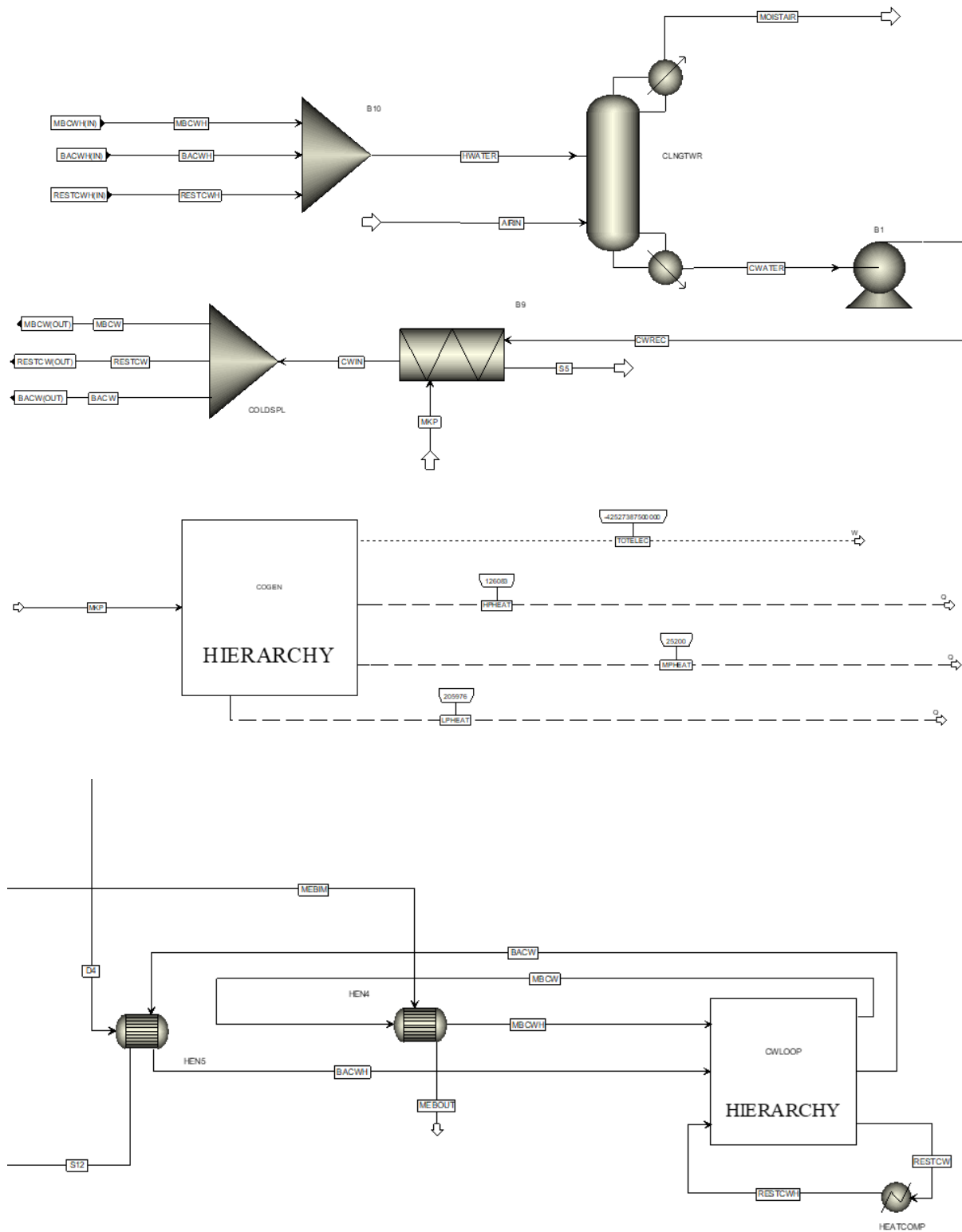
The horizontal streams are cold streams while vertical ones are hot streams.
 1, 2 and 3 are column distillate streams of column C2, C3, C4 respectively.
 Stream 5 is reactor inlet. Stream 4 is the reactor outlet that is fed into column C1.

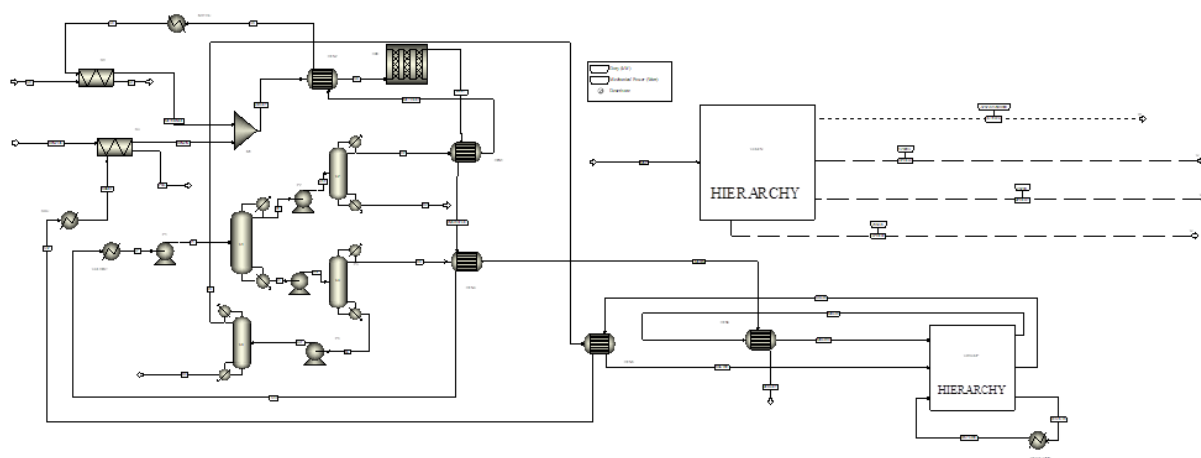
Aspen Flowsheet with Heat Exchanger Network: -



All heat exchangers have been adjusted with the minimum approach temperature of 10 °C.

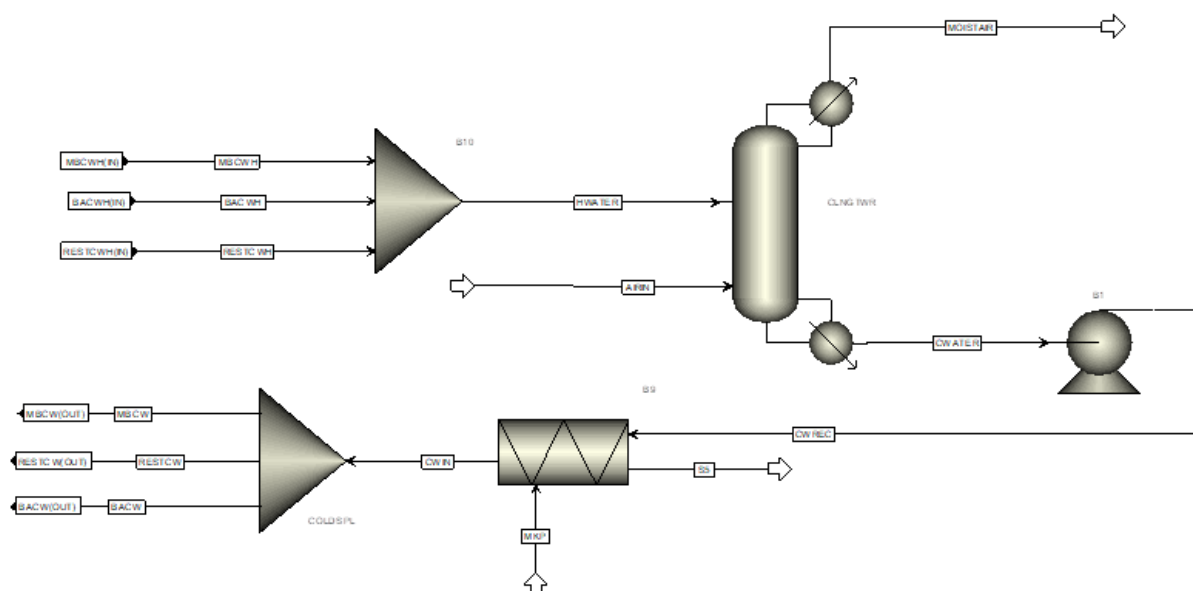
Overall Flow sheet (after correcting the utility processes):





Cooling Water Loop Design Improvement

In the previous report(report 3), we were not able to cool down the hot fresh water outlet of the heat exchanger (provided cool fresh water inlet). Now, we have managed the recycling in such a manner that it is being cooled and used as a fresh cool inlet, thus saving its cost.



Mass Balance in Column							
	MassIn (Air)	MassIn (Hwater)	MassIn_Total	MassOut (Cwater)	MassOut (Moistair)	MassOut_Total	
Methyl Benzoate	0	0	0	0	0	0	
Benzoic Acid	0	0	0	0	0	0	
WATER	42669.3037	11429947.23	11472616.53	10974439.2	498177.3311	11472616.53	
Methanol	0	0	0	0	0	0	
Phthalic Acid	0	0	0	0	0	0	
OXYGEN	3167237.29	3619.776521	3170857.068	3619.776538	3167237.292	3170857.069	
NITROGEN	10430899.7	8043.098427	10438942.8	8043.098464	10430899.71	10438942.8	
Total	13640806.3	11441610.1	25082416.4	10986102.07	14096314.33	25082416.4	

Stream Temperature and Pressure

	Air_in	Hwater	Cwater	Moist Air
Temperature (C)	25	39.98376	14.20964	30.75594
Pressure (bar)	1	1	0.84	0.8

Makeup Block

Mass Balance in Makeup Block				
	MassIn (CWREC)	MassIn (MKP)	MassIn_Total	MassOut (CWIN)
Methyl Benzoate	0	0	0	0
Benzoic Acid	0	0	0	0
WATER	10974439.2	455508.0282	11429947.22	11429947.23
Methanol	0	0	0	0
Pthalic Acid	0	0	0	0
OXYGEN	3619.776538	0	3619.776538	3619.776521
NITROGEN	8043.098464	0	8043.098464	8043.098427
Total	10986102.07	455508.0282	11441610.1	11441610.1

Design specifications utilized:

Design Spec	Manipulated variable	Target variable	Target value
1.	BACW Flow Fraction	BACWH Temperature	40 °C
2.	MBCW Flow Fraction	MBCWH Temperature	40 °C
3.	Total Mass Flow across MAKEUP	Total Fresh Cooling Water required	0 tonne/hr

Sensitivity Analysis For Profit Optimization:

We perform the sensitivity analysis to optimize the profit of our plant, in which we have included the below cost.

1. Reboiler Cost
2. Condenser Cost
3. Heater Cost
4. Tray Cost
5. Reactor Cost
6. Feed and Product stream Cost

We have developed a **Fortran Code** for the same and cost is attached below:

```
# QNET: total reboiler duty (sum of all column reboiler duties)
QNET = Q1 + Q2 + Q3 + Q4

# R_COST: reactor cost (based on reactor diameter and length)
R_COST = 17640 * (RDIA**1.066) * (RLENGTH**0.802)

# U: heat transfer coefficient
U = 0.852

# A1, A2, A3, A4: reactor heat transfer areas for each column
A1 = Q1 / (U * (527 - TEMP1))
A2 = Q2 / (U * (433 - TEMP2))
A3 = Q3 / (U * (527 - TEMP3))
A4 = Q4 / (U * (527 - TEMP4))

# CR_Ci: reactor cost for each reactor based on area
CR_C1 = 7296 * (A1**0.65)
CR_C2 = 7296 * (A2**0.65)
CR_C3 = 7296 * (A3**0.65)
CR_C4 = 7296 * (A4**0.65)

# REB_COST: total reboiler cost
REB_COST = CR_C1 + CR_C2 + CR_C3 + CR_C4
```

```
# U2: heat transfer coefficient (for condenser)
U2 = 0.568

# ACi: condenser area for each column, QCi = condenser duty
AC1 = QC1 / (U2 * (TEMPC1 - 298))
AC2 = QC2 / (U2 * (TEMPC2 - 298))
AC3 = QC3 / (U2 * (TEMPC3 - 298))
AC4 = QC4 / (U2 * (TEMPC4 - 298))

# CC_Ci: condenser cost for each condenser
CC_C1 = 7296 * (AC1**0.65)
CC_C2 = 7296 * (AC2**0.65)
CC_C3 = 7296 * (AC3**0.65)
CC_C4 = 7296 * (AC4**0.65)

# CC_TOT: total condenser cost
CC_TOT = CC_C1 + CC_C2 + CC_C3 + CC_C4

# H_COST: heater cost
H_COST = 25272 * HDUTY
```

```

# U2: heat transfer coefficient (for condenser)
U2 = 0.568

# ACi: condenser area for each column, QCi = condenser duty
AC1 = QC1 / (U2 * (TEMPC1 - 298))
AC2 = QC2 / (U2 * (TEMPC2 - 298))
AC3 = QC3 / (U2 * (TEMPC3 - 298))
AC4 = QC4 / (U2 * (TEMPC4 - 298))

# CC_Ci: condenser cost for each condenser
CC_C1 = 7296 * (AC1**0.65)
CC_C2 = 7296 * (AC2**0.65)
CC_C3 = 7296 * (AC3**0.65)
CC_C4 = 7296 * (AC4**0.65)

# CC_TOT: total condenser cost
CC_TOT = CC_C1 + CC_C2 + CC_C3 + CC_C4

# HCCOST: heater cost
HCCOST = 25272 * HDUTY

```

```

# LP1 and HP1: low-pressure and high-pressure steam costs
LP1 = 2.03 * 7200
HP1 = 5.66 * 7200

MEBCOST = 2200 * 7200
METHCOST = 370 * 7200
BENZCOST = 1010 * 7200
CWC = 0.378 * 7200

# MEBP, METHP, BENZP: material costs (MEB, Methanol, Benzene)
MEBP = MEBCOST * MEB
METHP = METHCOST * METH
BENZP = BENZCOST * BENZ

CWCP = (HXC1 + CXC + HXC2) * CWC

Q1_GJ = Q1 * 0.0036
Q2_GJ = Q2 * 0.0036
Q3_GJ = Q3 * 0.0036
Q4_GJ = Q4 * 0.0036

```

```

OPR1 = Q1_GJ * HP1
OPR2 = Q2_GJ * LP1
OPR3 = Q3_GJ * HP1
OPR4 = Q4_GJ * HP1

OPREB = OPR1 + OPR2 + OPR3 + OPR4

TRCOST = 19928246.1

TCC = CC_TOT + REB_COST + R_COST + TRCOST

YOC = H COST + OPREB + METHP + BENZP + CWCP

REV = MEBP

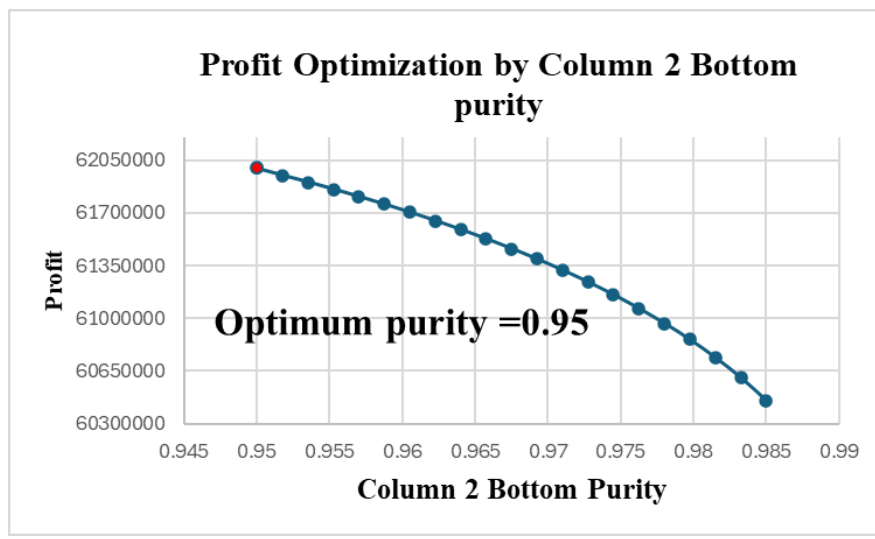
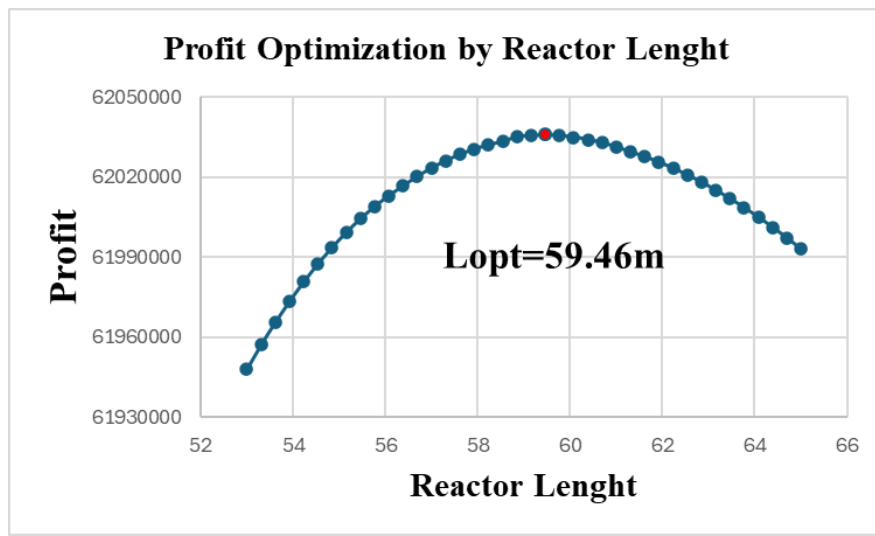
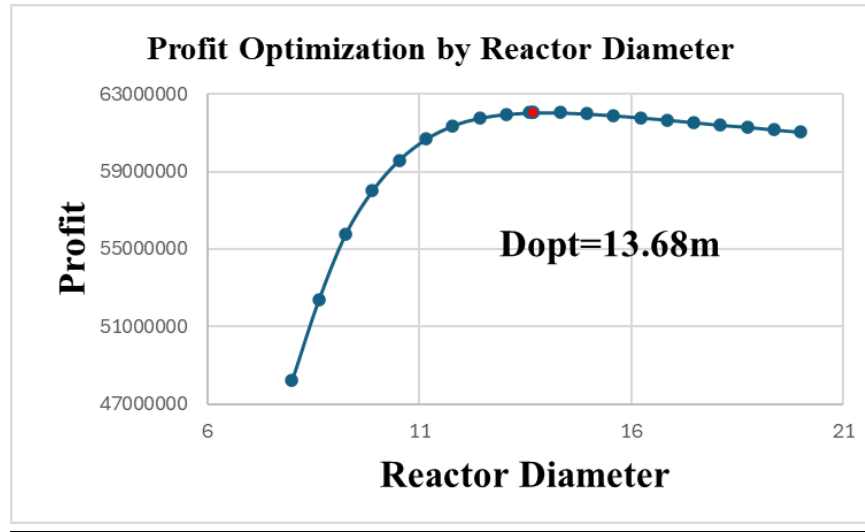
PROFIT = REV - (YOC) - ((TCC) / 3)

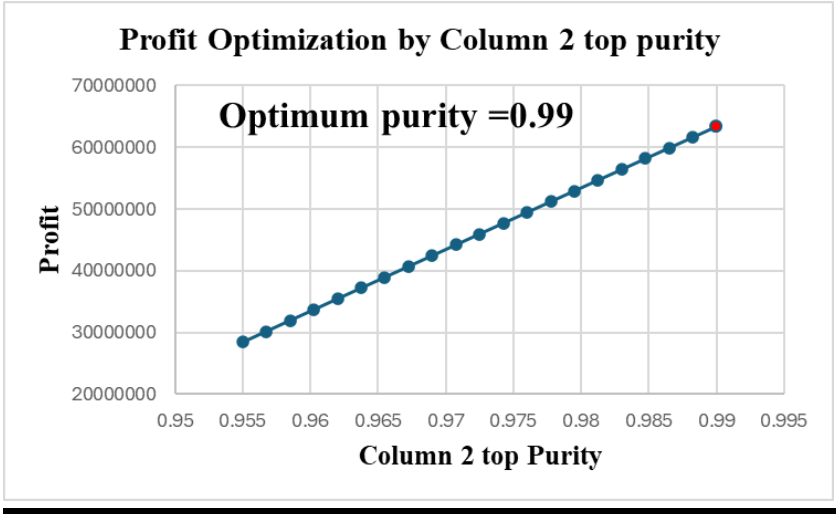
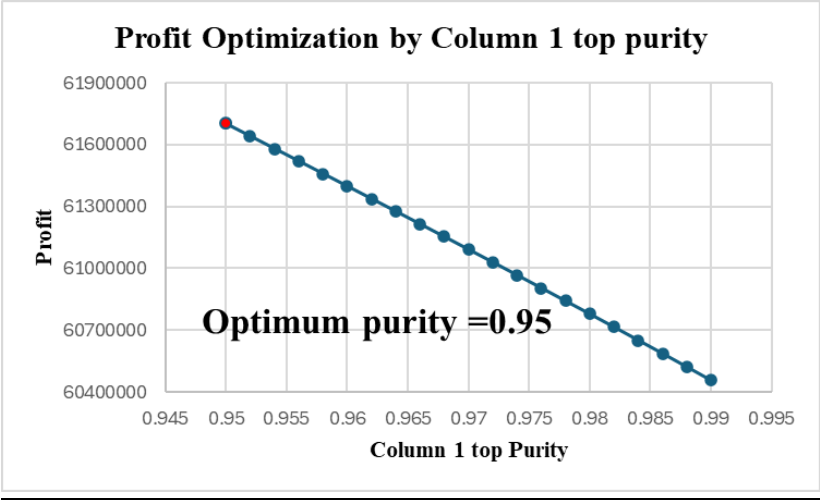
YR = REV - YOC

REV2 = REV - METHP - BENZP

```

We considered the reactor configuration (length and diameter), column 2 top and bottom product purity and column 1 top purity as dominant design variables. We performed sensitivity analysis by varying these parameters and found the optimum values for maximising our profit. We have limited the purities to not go below 95% in order to avoid the methanol wastage.





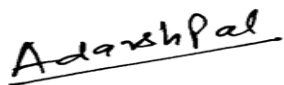
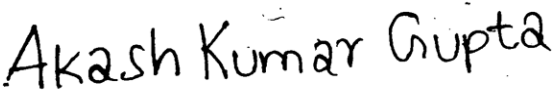
RESULTS:

TOTAL CAPITAL COST (TCC) (\$)	2.98×10^7
YEARLY OPERATING COST (YOC) (\$)	1.39×10^8
INPUT COST (RAW MATERIALS COST) (\$)	1.06×10^8
REVENUE (PRODUCT MARKET PRICE) (\$)	2.14×10^8
PAYBACK PERIOD (PBP) (years)	3
TOTAL ANNUALISED COST (TAC) (\$)	1.49×10^8
TOTAL ANNUAL PROFIT (TAP) (\$)	6.46×10^7

Objectives that could not be accomplished with reasons:

- The lack of information about the reaction kinetics and its behaviour at different temperatures has been an issue while integrating the reactor into the heat exchanger network.
- We have used the standard data for the heat exchanger design available on the internet. Also some settings of aspen could not be accommodated in the manual calculation of heat exchanger

Signatures of members:

Aaditya Amlan Panda	
Abhijit Dalai	
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Akash Kumar Gupta	
Kushagra Tiwari	
Saurabh Yadav	
Snehil Tripathi	
Tushar Verma	